

Yumeng He

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RESEARCH INTERESTS

My research interests lie in computer graphics and vision especially in physics-based animation, differentiable simulation, geometry processing, rendering, and 3D Gaussian Splatting.

EDUCATION

University of Southern California

Master of Science

Aug 2024 – May 2026

Los Angeles, CA, USA

- Cumulative GPA: 3.83/4.0
- Computer Science major
- Selected Coursework: Computer Graphics, Computer Animation and Simulation, 3D Graphics and Rendering

University of Toronto

Honours Bachelor of Science with High Distinction

Sep 2019 – May 2024

Toronto, ON, Canada

- Cumulative GPA: 3.71/4.0
- Computer Science and Mathematics double major
- Selected Coursework: Calculus I & II, Linear Algebra I & II, Ordinary Differential Equations, Abstract Mathematics, Complex Variables, Number Theory, Machine Learning, Artificial Intelligence

RESEARCH EXPERIENCE

UCLA AIVC Lab Visiting Student | Full-time

Supervisor: Prof. Chenfanfu Jiang

June 2025 – present

Los Angeles, CA, USA

- Conducting full-time research on 3D articulated object generation, with a focus on neural representations of shape and joint dynamics for physically plausible synthesis
- Exploring physics-aware scene layout and hierarchical composition, combining structural priors with simulation constraints to guide object placement
- Preparing a first-author submission to CVPR 2026, presenting a unified framework for articulated object synthesis guided by both data-driven learning and physical consistency

RELEVANT PROJECTS

Incompressible Fluid Simulation: A Comparison | C++, OpenGL

Spring 2025

- Designed and implemented four 2D incompressible fluid solvers—Stable Fluids, SPH, PIC, and APIC—from scratch in C++ with OpenGL-based real-time visualization
- Ran controlled experiments on identical test scenarios to evaluate visual accuracy, physical stability, and runtime performance
- Identified trade-offs between solver fidelity and speed; APIC achieved the most stable and detailed flows, while Stable Fluids offered the fastest performance with noticeable dissipation

Collision Detection with Penalty Method and IPC | C++

Spring 2025

- Built a physically accurate 3D jello cube simulator using a mass-spring system with structural, shear, and bend springs to model real-world deformation under force
- Implemented penalty-based collision detection to handle interactions with static obstacles—including inclined planes and spheres—ensuring realistic response under contact
- Upgraded basic collision logic with Incremental Potential Contact (IPC), reducing interpenetration to below $1e-4$ relative gap across 1,000+ simulation steps, effectively achieving zero visible penetration and stable behavior

Inverse Kinematics with Skinning | C++

Spring 2025

- Implemented Linear Blend Skinning and Dual Quaternion Skinning to deform 3D characters with smooth and realistic joint articulation
- Built a full FK–IK pipeline, supporting both forward kinematics for pose propagation and inverse kinematics for pose estimation under joint constraints
- Developed multiple IK solvers, including the pseudoinverse method (least squares) and Tikhonov regularization (damped least squares) to improve stability

Motion Capture Interpolation | C++

Spring 2025

- Implemented four interpolation schemes to reconstruct and smooth human motion sequences captured via optical motion capture
- Supported both Euler angles and quaternions for representing joint rotations, enabling robust handling of gimbal lock and interpolation artifacts
- Applied linear and Bezier interpolation across keyframes to generate natural, continuous transitions in high-dimensional skeletal animation

Ray Tracing | C++, Rendering

Fall 2024

- Built a modular ray tracer in C++ supporting both triangle meshes and spheres, featuring Phong shading, shadow rays, and recursive reflections for realistic mirror effects
- Integrated supersampling antialiasing, soft shadows via area lights, and the Moller-Trumbore algorithm for fast ray-triangle intersections
- Achieved visually convincing renders with enhanced realism and performance, balancing image quality and computational cost across complex scenes

Simulating a Roller Coaster | OpenGL, Rendering

Fall 2024

- Built a first-person roller coaster simulator using Catmull-Rom splines for smooth track generation and the Frenet frame for real-time camera orientation
- Engineered a multi-pass OpenGL rendering pipeline sustaining 120 FPS at 1080p, with realistic motion and responsive controls
- Added interactive features including user-controlled speed, auto-screenshot capture, and seamless on-rail camera transitions

Height Fields Using Shaders | OpenGL, Rendering

Fall 2024

- Developed a real-time 3D terrain visualization system using vertex and fragment shaders, rendering heightmaps with support for multiple modes: point cloud, line mesh, wireframe, and smooth shading
- Built a modular OpenGL pipeline with interactive controls for camera movement, zooming, rotation, and automated screenshot capture
- Integrated dynamic surface coloring based on elevation and lighting, enabling an immersive, high-performance terrain exploration experience

WORK EXPERIENCE

Software Developer Intern | Full-time

Aug 2022 – Aug 2023

HCL Canada Inc.

Toronto, ON, Canada

- Automated build and deployment pipelines using Jenkins, accelerating production releases and minimizing manual errors by eliminating key bottlenecks in CI/CD
- Containerized microservices for both x86_64 (xLinux) and IBM PowerPC (pLinux) architectures, enabling architecture-agnostic builds and scalable, consistent deployments
- Integrated SonarQube into the pipeline; documented a detailed usage guide, presented to cross-functional teams, and resolved 3 critical bugs and 308 code smells (including 4 blockers, 41 major issues)
- Deployed a shell script to auto-clean legacy logs weekly (Sunday 00:00 EST), reducing disk clutter and improving system performance with zero manual intervention

PROFESSIONAL SERVICES

- Mentor for Viterbi Graduate Mentorship Program (Fall 2025)

AWARDS & HONORS

- Dean's List Scholar for 4 years (2020 - 2024)
- Ranked in the top 25% of contestants in the Galois Contest (2017)
- Ranked in the top 25% of contestants in the Cayley Contest (2017)
- Ranked in the top 25% of contestants in the Canadian Intermediate Mathematics (2016)

TECHNICAL SKILLS

- **Operating Systems:** Linux, Windows, MacOS
- **Programming Languages:** C++, Python, JavaScript, HTML, CSS, MySQL
- **Graphics and Simulation:** OpenGL, GLSL, Eigen, PyTorch
- **Tools and DevOps:** Docker, Jenkins, Gradle, Ant, Shell, Git, Github Actions
- **Languages:** Mandarin (native), English (fluent)